Skill-4&5

Name: J. Simran

Rollno:2520030166

Step-1: Insert Customer Data

Customer 1:

{

  "customer_id": 101,

  "name": "Ravi Kumar",

  "city": "Hyderabad",

  "account_type": "Savings",

  "balance": 85000,

  "status": "Active"

}

Customer 2

{

  "customer_id": 102,

  "name": "Priya Sharma",

  "city": "Vijayawada",

  "account_type": "Current",

  "balance": 125000,

  "status": "Active"

}

Customer 3

{

  "customer_id": 103,

"name": "Arjun Reddy",

"city": "Hyderabad",

"account_type": "Savings",

"balance": 45000,

"status": "Active"

}

Customer 4

{

"customer_id": 104,

"name": "Sneha Rao",

"city": "Warangal",

"account_type": "Savings",

"balance": 210000,

"status": "Active"

}

Customer 5

{

"customer_id": 105,

"name": "Kiran Kumar",

"city": "Guntur",

"account_type": "Current",

"balance": 65000,

"status": "Inactive"

}

Customer 6

{
  "customer_id": 106,
  "name": "Anjali Singh",
  "city": "Hyderabad",
  "account_type": "Savings",
  "balance": 175000,
  "status": "Active"
}

Customer 7

{
  "customer_id": 107,
  "name": "Rahul Verma",
  "city": "Bengaluru",
  "account_type": "Current",
  "balance": 95000,
  "status": "Active"
}

Customer 8

{
  "customer_id": 108,
  "name": "Meena Devi",
  "city": "Chennai",
  "account_type": "Savings",
  "balance": 320000,
  "status": "Active"

```
}
```

Step-2: Insert Transaction Data

Transaction 1

```
{
  "transaction_id": 1001,
  "customer_id": 101,
  "type": "Deposit",
  "amount": 25000,
  "mode": "UPI",
  "date": "2026-08-01",
  "status": "Success"
}
```

Transaction 2

```
{
  "transaction_id": 1002,
  "customer_id": 101,
  "type": "Withdrawal",
  "amount": 10000,
  "mode": "ATM",
  "date": "2026-08-02",
  "status": "Success"
}
```

Transaction 3

```
{
  "transaction_id": 1003,
```

"customer_id": 102,
"type": "Deposit",
"amount": 50000,
"mode": "NEFT",
"date": "2026-08-02",
"status": "Success"
}

Transaction 4

{
"transaction_id": 1004,
"customer_id": 102,
"type": "Withdrawal",
"amount": 15000,
"mode": "ATM",
"date": "2026-08-03",
"status": "Success"
}

Transaction 5

{
"transaction_id": 1005,
"customer_id": 103,
"type": "Deposit",
"amount": 30000,
"mode": "UPI",
"date": "2026-08-03",

"status": "Success"

}

Transaction 6

{

"transaction_id": 1006,

"customer_id": 103,

"type": "Withdrawal",

"amount": 5000,

"mode": "ATM",

"date": "2026-08-04",

"status": "Success"

}

Transaction 7

{

"transaction_id": 1007,

"customer_id": 104,

"type": "Deposit",

"amount": 100000,

"mode": "NEFT",

"date": "2026-08-04",

"status": "Success"

}

Transaction 8

{

"transaction_id": 1008,

"customer_id": 104,
"type": "Withdrawal",
"amount": 25000,
"mode": "ATM",
"date": "2026-08-05",
"status": "Success"
}

Transaction 9

{
"transaction_id": 1009,
"customer_id": 105,
"type": "Deposit",
"amount": 15000,
"mode": "UPI",
"date": "2026-08-05",
"status": "Success"
}

Transaction 10

{
"transaction_id": 1010,
"customer_id": 106,
"type": "Deposit",
"amount": 75000,
"mode": "NEFT",
"date": "2026-08-06",

"status": "Success"
}

Transaction 11

{
"transaction_id": 1011,
"customer_id": 106,
"type": "Withdrawal",
"amount": 20000,
"mode": "ATM",
"date": "2026-08-06",
"status": "Success"
}

Transaction 12

{
"transaction_id": 1012,
"customer_id": 107,
"type": "Deposit",
"amount": 40000,
"mode": "IMPS",
"date": "2026-08-07",
"status": "Success"
}

Transaction 13

{
"transaction_id": 1013,

"customer_id": 107,
"type": "Withdrawal",
"amount": 12000,
"mode": "ATM",
"date": "2026-08-07",
"status": "Success"
}

Transaction 14

{
"transaction_id": 1014,
"customer_id": 108,
"type": "Deposit",
"amount": 150000,
"mode": "NEFT",
"date": "2026-08-08",
"status": "Success"
}

Transaction 15

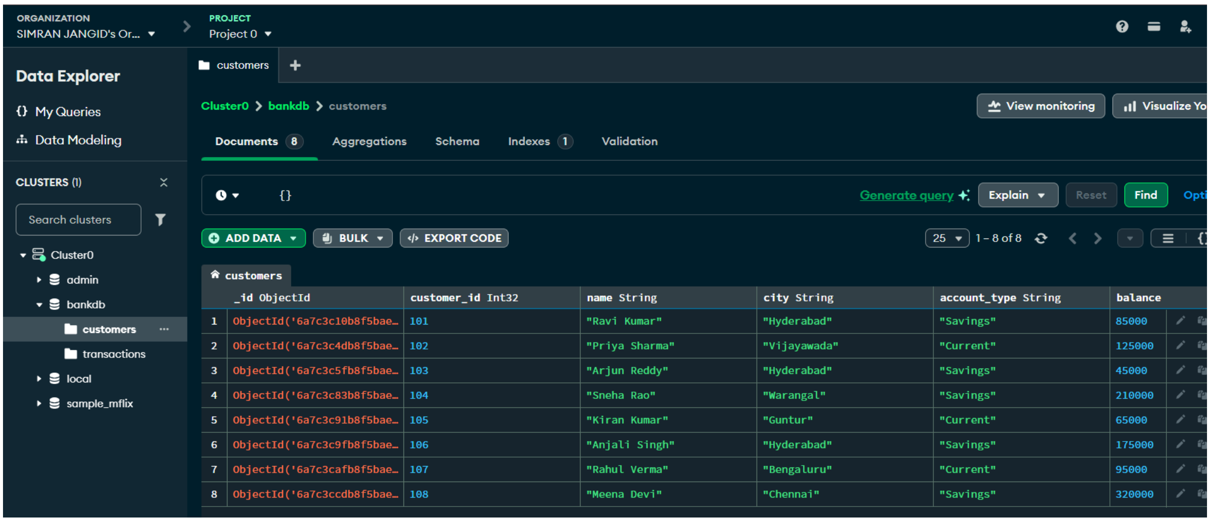
{
"transaction_id": 1015,
"customer_id": 108,
"type": "Withdrawal",
"amount": 30000,
"mode": "ATM",
"date": "2026-08-08",

"status": "Success"

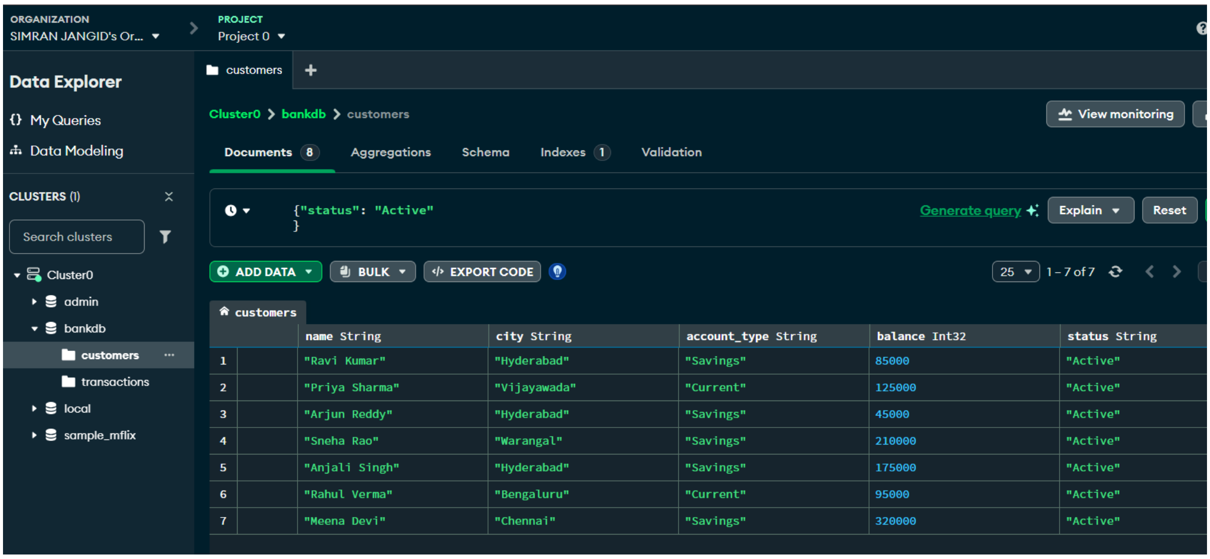
}

Step-3: Execute the Queries:

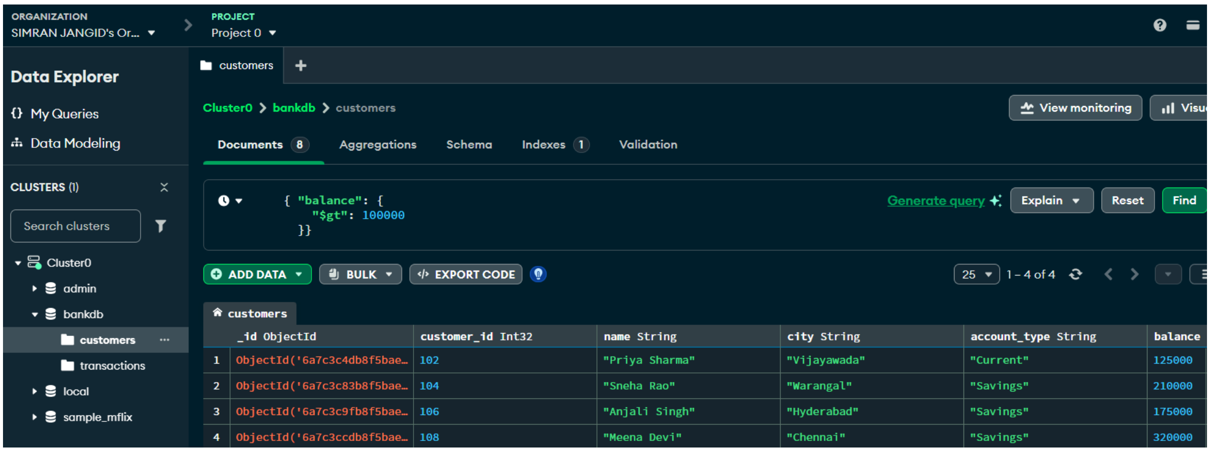
Query 1 — Display All Customers



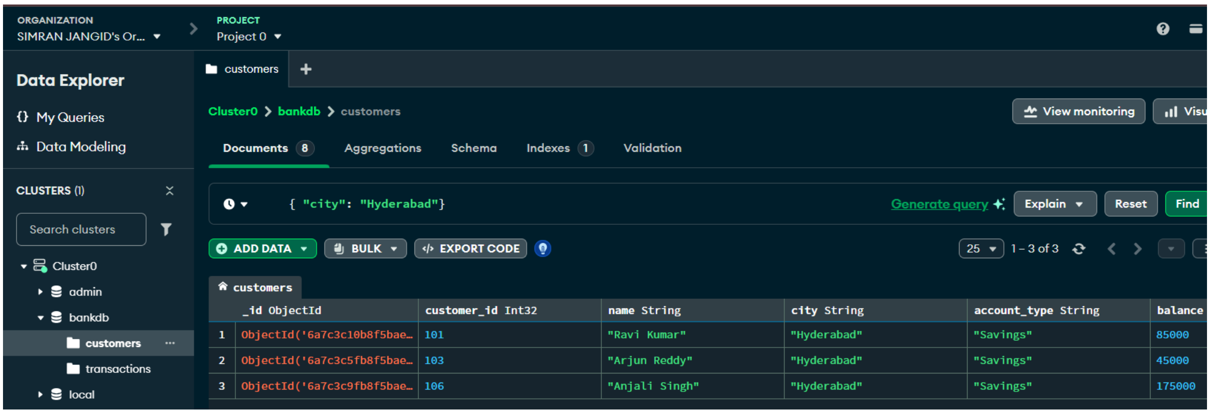
Query 2 — Find Active Customers



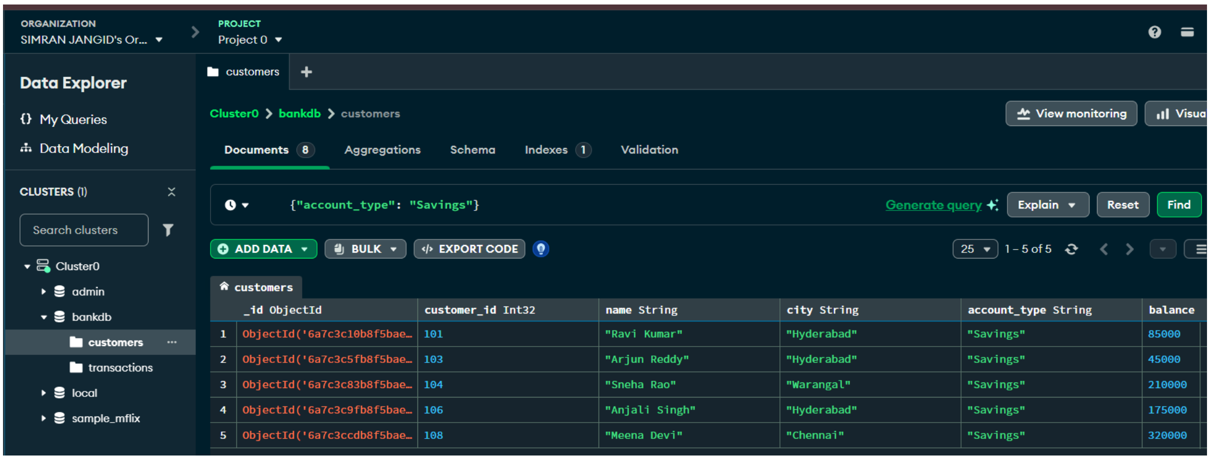
Query 3 — Balance Greater Than 100000



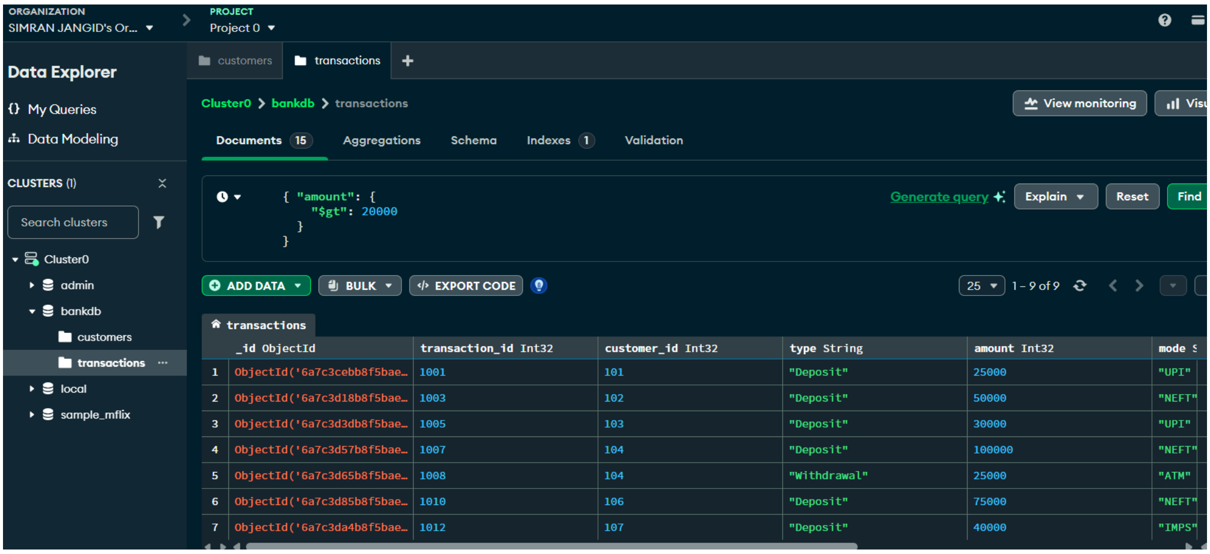
Query 4 — Hyderabad Customers



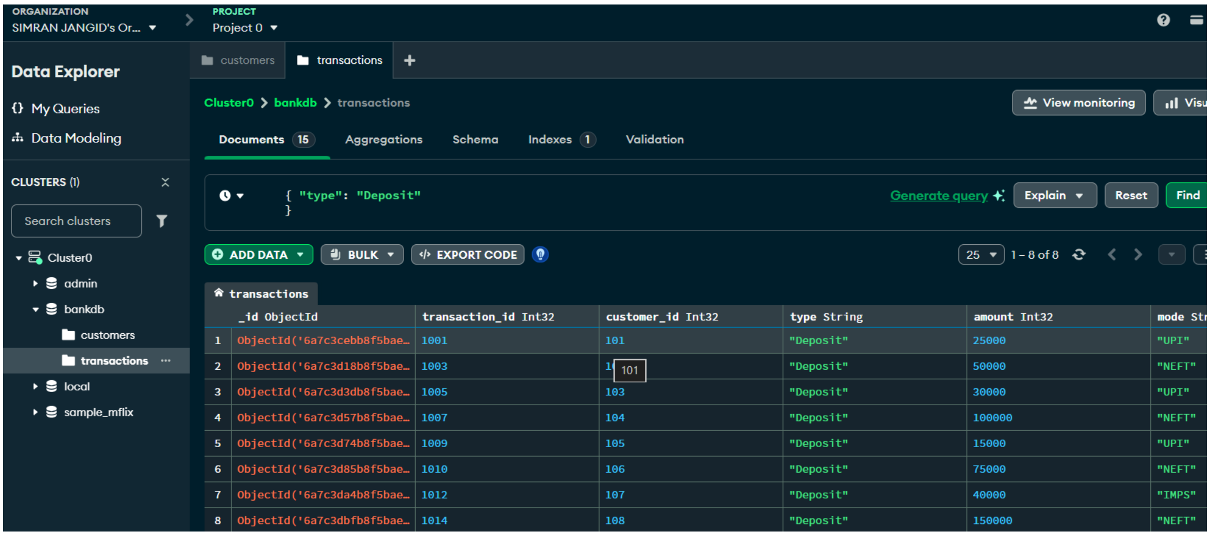
Query 5 — Savings Account Customers



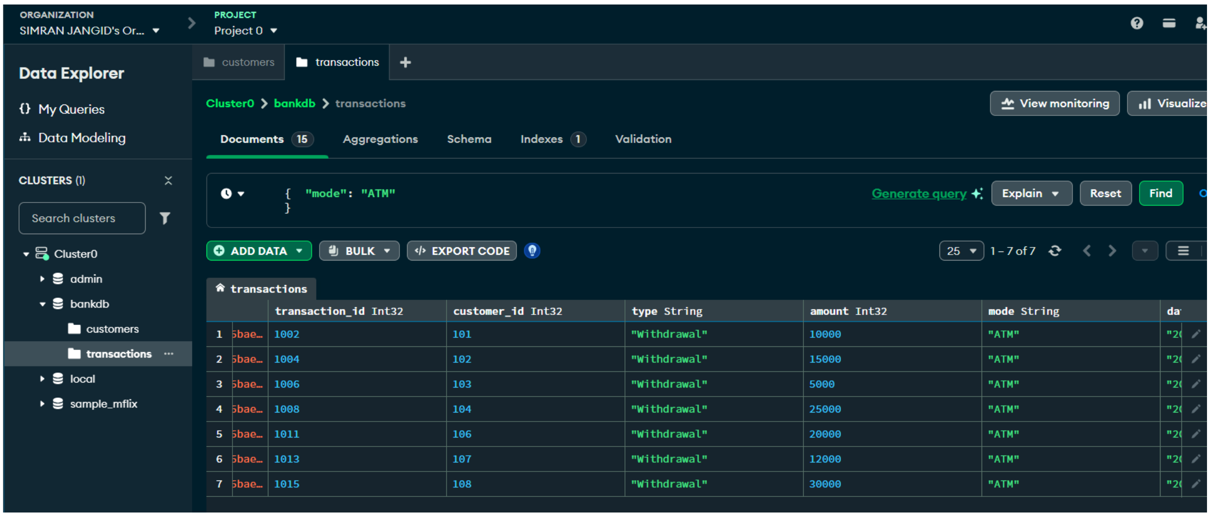
Query 6 — Transactions Greater Than 20000



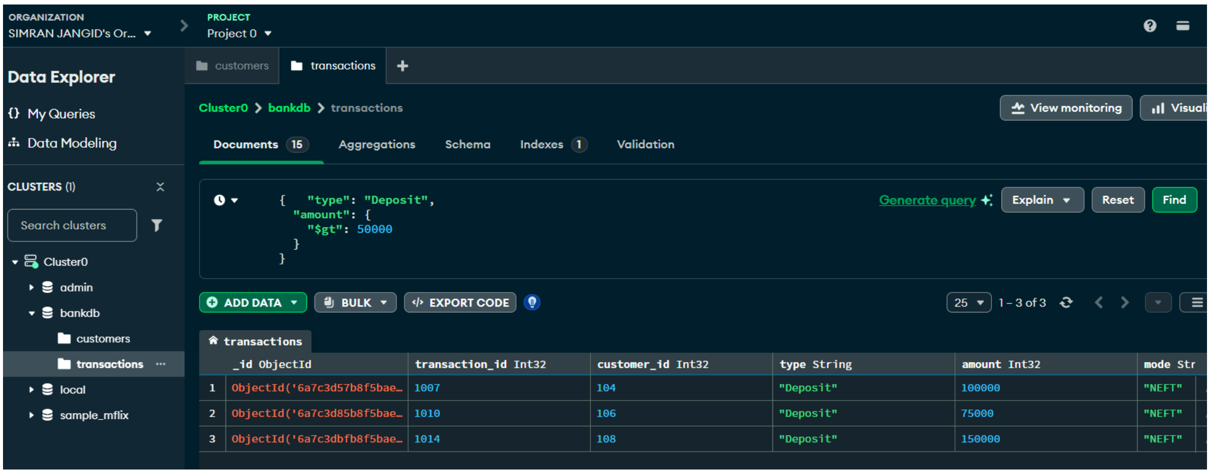
Query 7 — Find Deposits



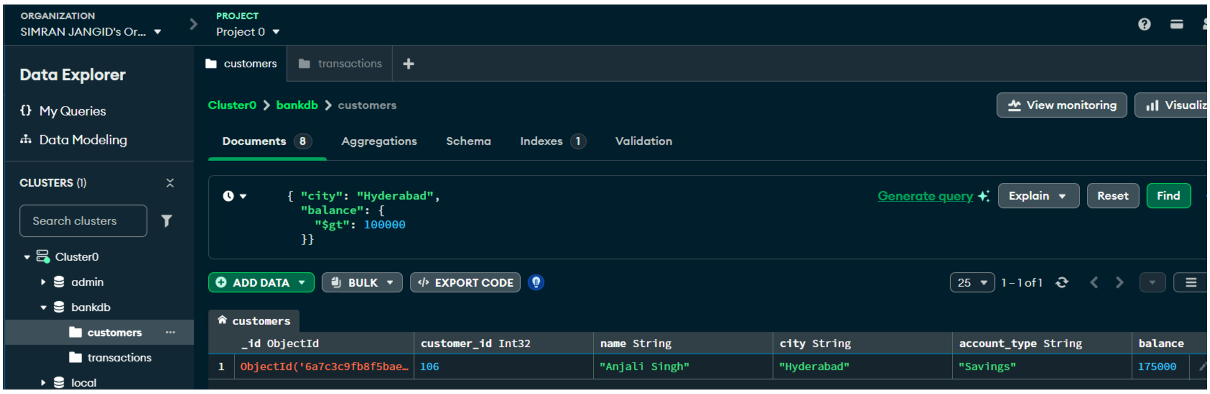
Query 8 — Find ATM Transactions



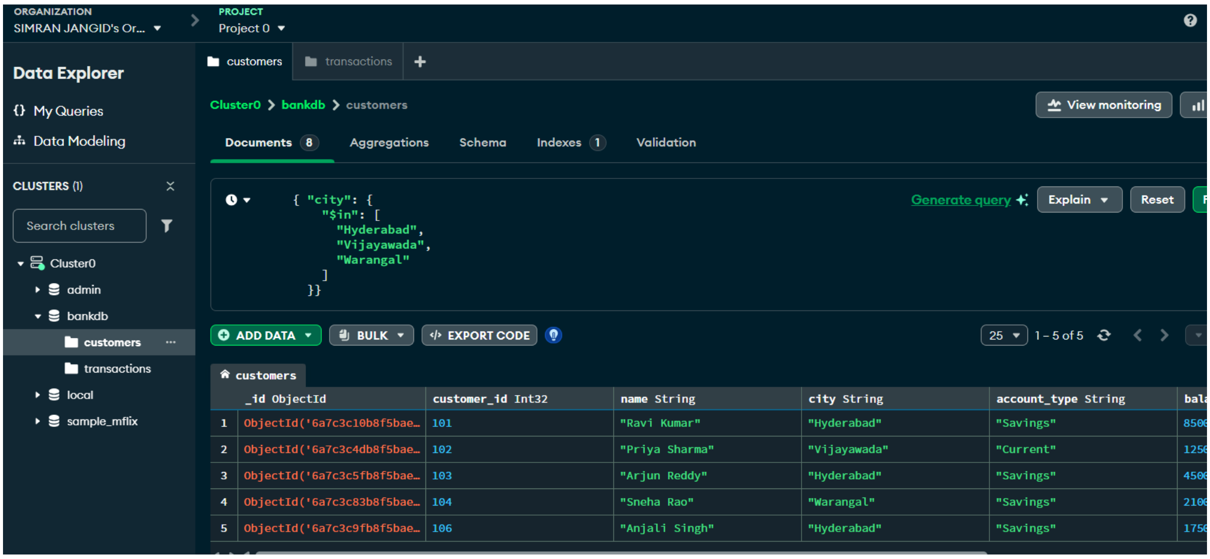
Query 9 — Deposits Greater Than 50000



Query 10 — Hyderabad Customers with Balance > 100000

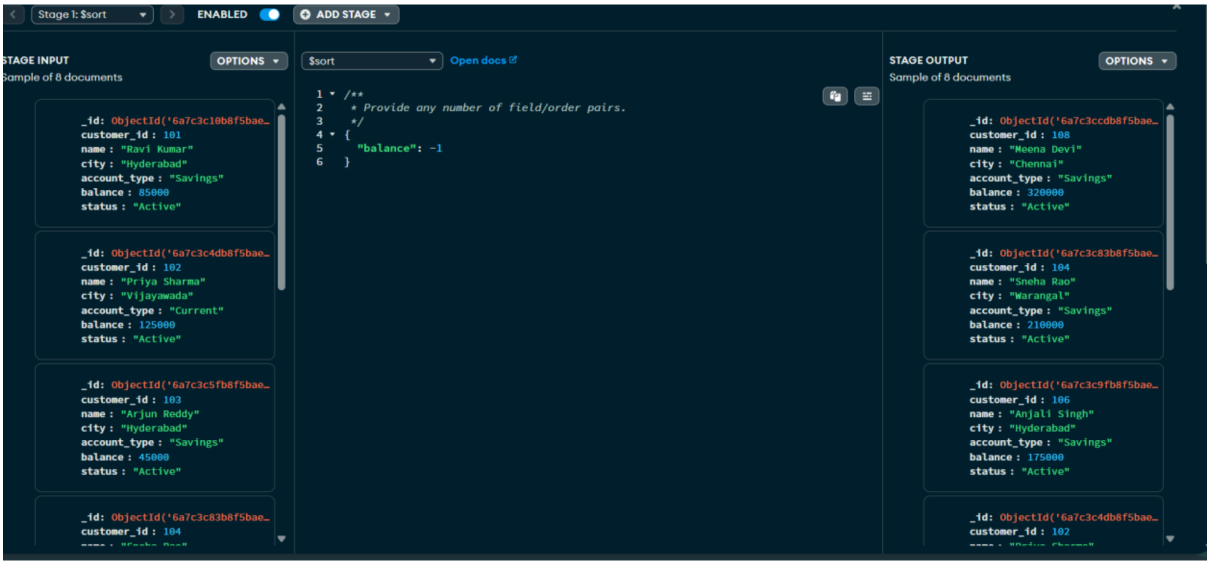


Query 11 — Customers From Multiple Cities

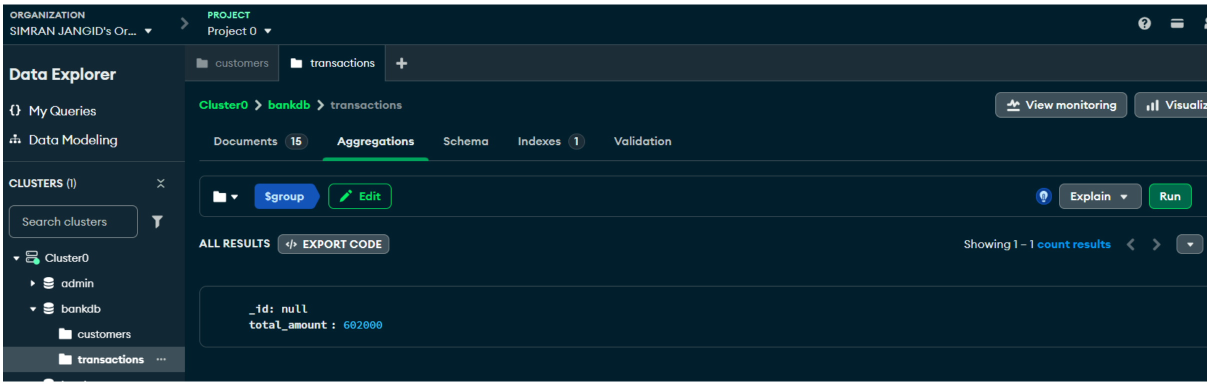


Step-4: Aggregation Queries

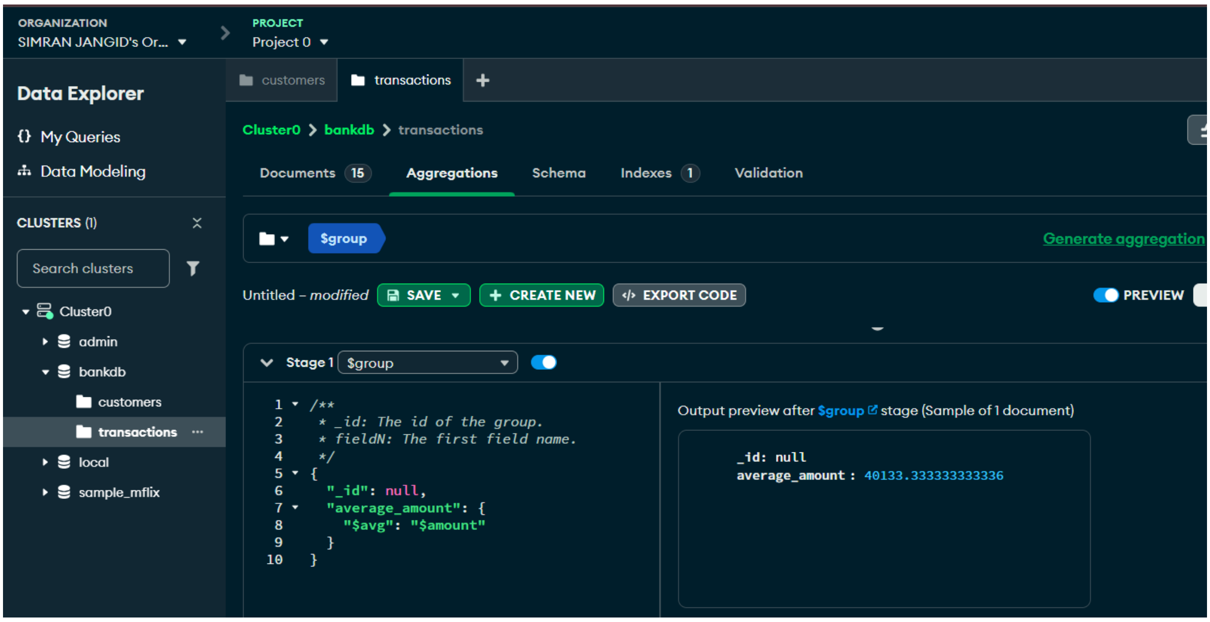
Query 12 — Sort Customers by Balance



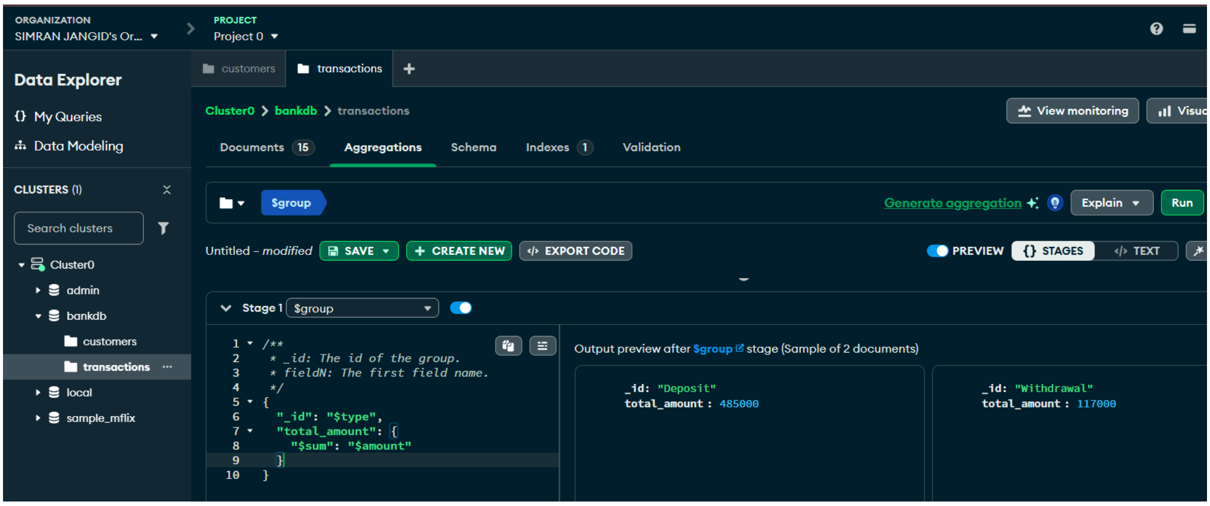
Query 13 — Total Transaction Amount



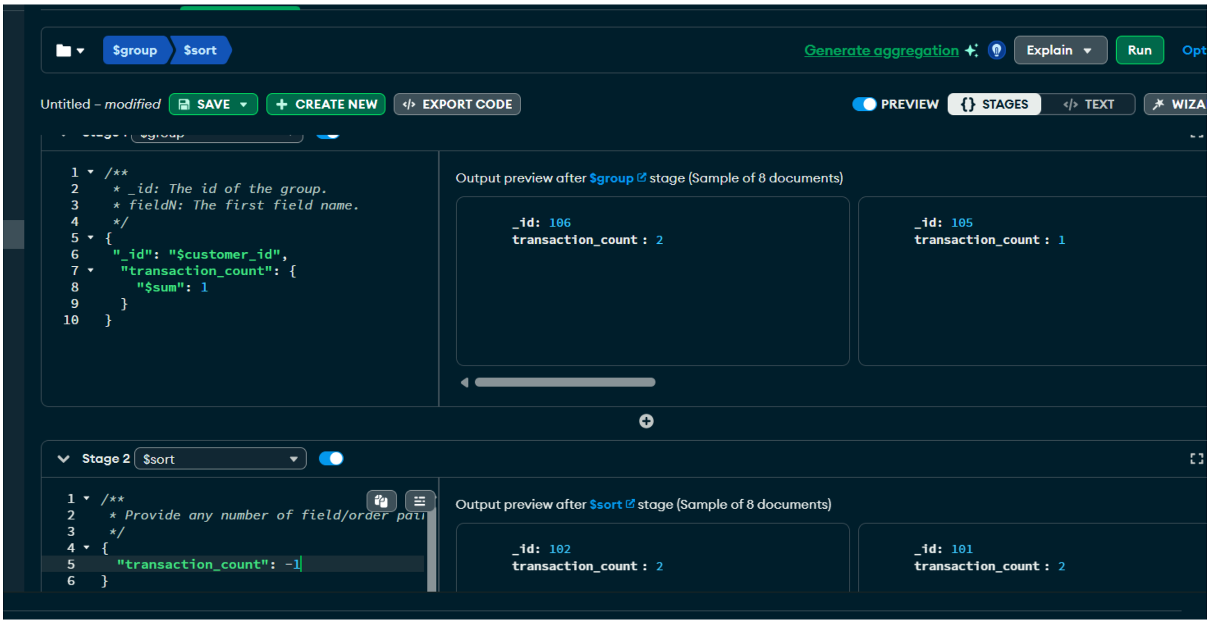
Query 14 — Average Transaction Amount



Query 15 — Total Amount by Transaction Type



Query 16 — Number of Transactions per Customer


$group
$sort
Generate aggregation
Explain
Run
Untitled – modified
SAVE
CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
/**
* _id: The id of the group.
* fieldN: The first field name.
*/
{
"_id": "$customer_id",
"transaction_count": {
"$sum": 1
}
}
Output preview after $group stage (Sample of 8 documents)
_id: 106
transaction_count : 2
_id: 105
transaction_count : 1
Stage 2
$sort
/**
* Provide any number of field/order pair
*/
{
"transaction_count": -1
}
Output preview after $sort stage (Sample of 8 documents)
_id: 102
transaction_count : 2
_id: 101
transaction_count : 2